MIT 18.06SC - 2.1

Problem 16.1

Let $y_1 = y_2 = 1$ and $y_3 = -1$.
Then we have:-

$$\begin{array}{rcrcrcl} x_1 & - & x_2 & & & = & 1 \\ & & x_2 & - & x_3 & = & 1 \\ -x_1 & & & + & x_3 & = & -1 \\ \hline & & & & 0 & = & 1 \end{array}$$

Problem 16.2

a) For r to be a basis for $C(A)$ and n to be a basis for $N(A^T)$, we need $r \perp n$. ~~Similarly~~ Also, for l to be a basis for $N(A)$ and c to be a basis for $C(A^T)$, we need $l \perp c$.

Also, A must have rank 1, and $\dim(N(A)) + \dim(C(A^T))$ must be n, similarly we must have $\dim(N(A^T)) + \dim(C(A)) = m$.

b) Let $A = rc^T$, then each column is a multiple of r so r is a basis for the column space, and each row is a multiple of c so c is a basis for the row space. ~~Since~~